

Condensed Morse Code: An alternative writing system for Morse code designed for learners

Sami A. Sued

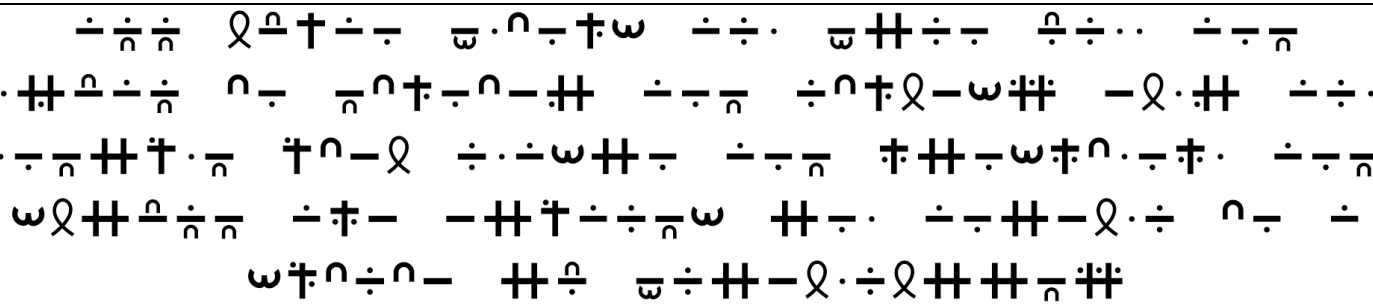


Fig 1. Article 1 of the Universal Declaration of Human Rights (United Nations, 2025) in Condensed Morse Code.

Abstract

This report presents a more space-efficient featural writing system for international Morse code called Condensed Morse Code (CMC) that is geared towards ease of reading and writing for humans, intended to be used as a learning tool for beginners. It considers previous efforts, as well as varying aspects of writing systems and neographies for a planned approach to the design process, with the drafting and development of CMC actively shown. The result provides a familiar format for preexisting Morse code users compact in space, comprised of visually distinct glyphs, optimised for quick handwriting, with a standard set of rules that eliminates ambiguity. It has limitations in its ability to be used in other standards for Morse code and languages other than English, though it can be adapted to suit those needs in further iterations.

Keywords: Morse code, writing systems, neography, script design, learning tools for beginners

Introduction

Morse code, a system that uses short and long signal durations (“dits” and “dahs”) to encode text has always been written using dots and dashes (· and -) as representations of the two signal durations. This has remained the standard method for transcribing it throughout its usage, and although Morse code is less used now than it was in the early 20th century due to the invention of more effective telecommunication methods, it is still used by amateur radio hobbyists and Morse code enthusiasts.

As a Morse code beginner, however, I found that Morse code often took long to transcribe, was difficult to read in its written form, and often resulted in high usage of horizontal page space, while verticality was essentially unused. For that reason, I investigated the potential of making a more compact alternative to writing individual dits and dahs (which I shall refer to as “units”) sequentially, which would aid learners in quickly writing and intuitively reading it while requiring minimal learning time to pick up. Such an alternative would require a digital font as well, as it would allow learners to convert any text to its form, making active transmission of messages written in it potentially easier when read due to pattern recognition. This would also be useful as a learning tool for transcribing Morse code compactly.

During research of prior work on the topic, I discovered a similar project (and the only other remotely formalised one of its kind) called Compact Morse Code (Lapcionak, 2025). It achieves the goal of creating a more compact writing system for Morse code, with evidence of a handwritten alternative form of it which uses abbreviations to represent unit combinations. However, there are a set of issues with Compact Morse Code that make it unfit for my aims:

- Undocumented handwritten form and conventions
- Multiple forms rather than a unified one
- No license, making it an “all rights reserved” work by default
- Overly reliant on vertical stacking of characters, which can make its printed form harder to read, while making the handwritten form utilise unique symbols unfamiliar to Morse code users

- Differentiates between capital and lowercase letters, which is a feature not present in Morse code

These shortcomings justified further development of a new writing system not based on Compact Morse Code, with its goals as follows:

1. Intuitive for current users of Morse code to understand
2. Significantly more compact than individual dots and dashes (must fit in traditional ruled paper)
3. The different representations of characters must be visually discernible from each other
4. Easily and quickly handwritten
5. Clearly documented conventions of usage in one unified form

Approach

The construction of a neography (a writing system constructed artificially rather than having evolved over time as a natural script) for Morse code must be done in a structured approach, such that design choices are made rationally and are justifiable. One important thing to note is that due to the existence of multiple standardised versions of Morse code, this writing system will be focused on international Morse code (International Telecommunications Union, 2009).

The primary goal of the writing system is to be intuitive to users accustomed to the traditional representation of Morse code. A writing system can be thought of as intuitive if it requires little to no learning time to understand given a specific background. As such it should ideally bear some resemblance to the traditional writing system. The neography doesn't need to reform the individual characters used to represent the units of Morse code, as is done sometimes when creating neographies (such as with the case of Turkish switching from its adapted version of the Arabic abjad to the Latin alphabet which better represented its phonemes), as replacing the dot and dash as the foundational representations of the units would result in a less intuitive writing system to preexisting users.

As such, the task at hand is instead to find a way to intuitively lay out the dots and dashes in the most optimal way that compromises between all the other goals of the writing system. Compromise is specifically emphasised because there is no way to perfectly capture all the goals simultaneously without conflict. For example, maximal spatial efficiency would likely conflict with the goal of

visual discernibility, as tightly compressed characters can often “blur” into each other. Narrowing down what type of writing system best achieves this is the first step to designing one. Since Morse code is simply a way of transmitting information in any language, thinking of it as if it were a replacement for the phonetics of a language can thus be a helpful analogy in this process.

From that point of view, units can be thought of as the two “phonetic features” (the basic unit of phonological structure that differentiates sounds apart) of the language. Following this logic, a set of units creates a “phoneme”, which in this case would be a character, such as a letter or a number. While in traditional transcription of Morse code each dot and dash is written as its own individual glyph, a more compact system can be formulated by using glyphs that group the writings of “phonemes” such as letters, just like in the alphabet. This must be done with intuitiveness in mind to ensure intelligibility for preexisting users. The way the dots and dashes are arranged must follow a consistent and simple ruleset where the intended order of the set is clear.

There are six types of writing systems (Daniels, 2000):

- Logosyllabary: Each character stands for a morpheme
- Syllabary: Each character stands for a syllable
- Abjad: Each character stands for a consonant
- Alphabet: Each character stands for a consonant or vowel
- Abugida: Each character stands for a consonant accompanied by a particular vowel, which can be changed by adding to the character
- Featural Script: Shapes of characters correlate with phonetic features

Since the dots and the dashes are to be considered as “phonetic features”, only the featural script which considers phonetic features applies to this scenario. It can thus be concluded that when assuming that Morse code is a replacement for the phonetics of a language, the most intuitive writing system would be a featural one.

The next step to developing this writing system is deciding the directionality and orientation of the script and the glyphs. Given that the most widely used version of Morse code is in English, the glyphs should be arranged in a left-to-right (hereafter LTR), top to bottom direction, just like in English. It is much more difficult to choose the way in which to arrange the dots and dashes of Morse code

representations of letters in their respective graphemes in such a way that makes the order clear.

The first important feature is that due to the length of dashes, the empty space above or below them is not only available, but clearly separate, making them more suitable to be the “centres” of a writing system, where the reading order of the units is implied by their placement respective to the dash, or lack of it. This dash-centred approach where dots accompany dashes leaves two ways of arranging said dashes: vertically or horizontally. Although the vertical stacking approach is more compact, longer morse representations can take up too much vertical space to fit in ruled paper if the dots and dashes are stacked, as seen in the case of Compact Morse Code. It is instead more intuitive to preexisting users for dashes to be put side-to-side, which is also more fitting to ruled paper, since horizontal space is ubiquitous.

Thus, the only remaining issue is on the representation of multiple dashes in one grapheme horizontally while showing clear separation. The solution devised for CMC is for all the varieties of dash graphemes to centre around a horizontal line, with each variety using increasing numbers of vertical strokes to separate the horizontal line into “dash sections”. If there is only one dash in a letter, the line needs no separations. Whenever there are no dashes, the dots are best simply written as a single grouped unit. This sets out the basis for the first draft of the writing system, shown in figure 2.



Fig 2. First draft of Condensed Morse Code

From this first draft, there is already a clear improvement in compactness, as well as some clear issues with discernibility and ease of handwriting. The pairing of two dots that corresponds to “I” is very similar to two consecutive “E”s, one dot each. This, while potentially solvable by vertically stacking dots as in Compact Morse Code, is a poor approach when considering the case of several dots preceding a dash, such as the Morse code for “4” (····-), as the vertical space would be insufficient if the dash is kept in the middle. Moving the dash away from the middle risks it approaching the page’s line, which would result in it potentially being interpreted as four dots instead (the Morse code for “H”) if the dash merges with the page line.

Using individual graphemes to represent consecutive dot combinations is a better approach, because it makes dot combinations quickly recognisable if done correctly, and does not result in a loss of intuitiveness if the graphemes are designed with an underlying logic. They must be quick to write and must look visually distinct, even if written quickly, so the aim is to ideally need one stroke to write a dot combination grapheme, making it quicker to write neatly than multiple dots. An example of this practice can be found in Arabic handwriting, where multiple dots (i’jam) are replaced with similar symbols, shown in figure 11 (Ullah & Jamjoom, 2022). The graphemes that were chosen to represent dot combinations (in figure 3), as well as their explanations (in figure 4) are shown.

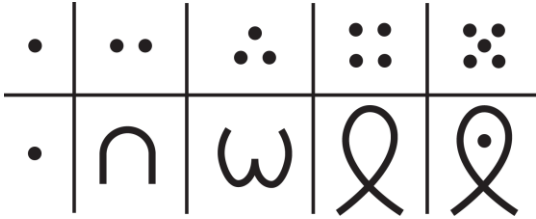


Fig 3. Dot combination graphemes

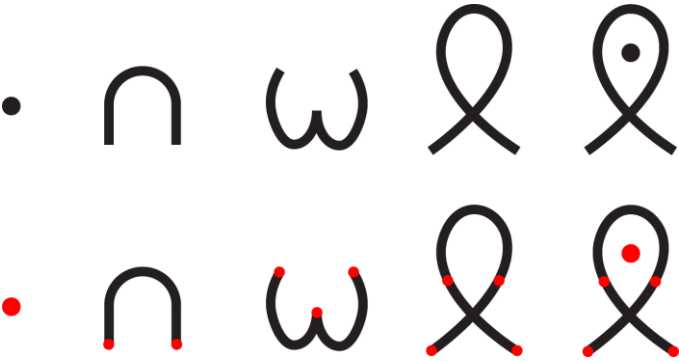


Fig 4. Explanation behind grapheme designs

Each grapheme uses one stroke, except for the grapheme representing five consecutive dots, which is only used for the Morse code representation of “5”. This set of graphemes ensures that there is no combination of consecutive dots in ITU international Morse code that cannot be written with one grapheme, except for the prosign for an error in a message or transmission, which has no reason to be transcribed or written in advance. This addition forms the second draft of CMC. An example is shown in figure 5.



Fig 5. Second draft of Condensed Morse Code.

With the second draft, many of the issues of the first have been addressed. There are a few more steps left in the formalisation of CMC before it can be converted into a font. First is the largely latent issue of alternative writings (allographs) for a single English letter when multiple dash sections are present. For example, the letter “K” in Morse code can be written in two ways, shown in figure 6.



Fig 6. Allographs for “K” in Condensed Morse Code

This becomes an even bigger issue for increasing numbers of dashes, where the number of allographs can grow significantly. Though for handwriting this can be acceptable at best, and cumbersome at worst, it makes a conversion to font form where a letter has a single glyph associated with it impossible.

Thus, the two approaches to address this are either to pick individual glyphs for each letter as the “correct” ones with leeway for allographs in handwriting, or to set out a set of guidelines that can be used to infer which glyph from the possible allographs is the preferred writing. The first approach is both labour-intensive (due to having to manually pick a “correct” choice for each allograph-susceptible glyph) and can result in arbitrary choices, which might themselves be subconsciously led by “unwritten” guidelines. This makes the second approach significantly more rational, while also making justified choices possible.

The choice of guidelines to infer glyphs is largely a matter of aesthetics. Something often seen as aesthetically pleasing is the property of symmetry (Hübner & Thömmes, 2019), so it was picked as the main priority when dealing with such cases. Most of the time however, symmetry is not achievable, so a reasonable rule to adopt is to favour dot grapheme placement on the bottom half of a dash, as letters in writing tend to gravitate towards the page line below them. Thus, the “allograph resolution rules” are as follows:

In case of allographs:

- If there is an allograph which achieves reflectional symmetry, it is the preferred writing.
- If not, the allograph whose dot graphemes lean towards the bottom left dash sections is preferred (as opposed to their alternate position in the following dash section).

With that, one last addition that naturally arose when experimenting with the calligraphy of CMC was that of changing the layout of the 2-dash grapheme from the shape of a plus into the shape of a cross with a narrower dash (as shown in figure 7). This was done for three reasons:

- It provides more vertical space at the bottom dash sections since dash sections are less likely to contain characters on top of them, as well as providing more horizontal space for further graphemes.
- It makes characters with two dashes distinct from those that are not, which makes it easier to recognize symbols, due to the visible height difference. This is especially important because around a third of the symbols in international Morse code fit the 2-dash pattern.
- The cross symbol’s previously stated height difference also serves an aesthetical purpose, as it provides more variety in the shape of characters.

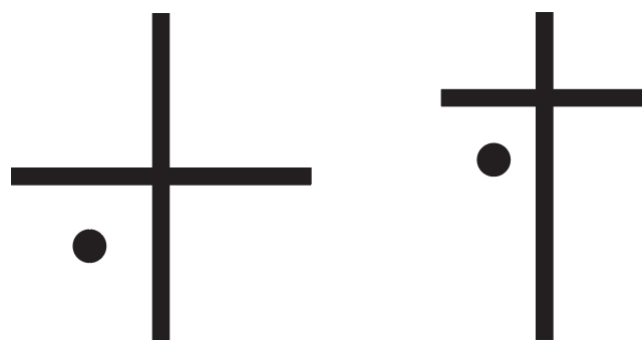


Fig 7. The old and new writings of the glyph for “K”

Thus, these changes form the finished version of CMC. The writing system has been formalised. The standard of CMC is as follows:

1. In CMC, Morse code representations of characters are replaced by glyphs comprised of graphemes.
2. The two types of graphemes are “dot graphemes” and “dash graphemes”, where dash graphemes are vertically centred with respect to the row.

3. Glyphs are arranged identically to letters in English (left-to-right) and utilise the same spacing between words.
4. Multiple dashes are represented by a horizontal line split into several “dash sections” using vertical strokes, where the number of dash sections in a dash grapheme is proportional to the number of dashes.
5. Dots that occur before dashes are placed on top of the dash sections, while dots after dashes are placed below.
6. Multiple consecutive dots are abbreviated with dot combination graphemes (up to five consecutive dots).
7. If there are multiple allographs of a glyph, the correct writing is chosen by the rules below:
 - a. If there is an allograph which achieves reflectional symmetry, that is the correct writing.
 - b. If not, the allograph whose dot graphemes lean towards the bottom left dash sections is preferred.
8. The horizontal line in rule four used for dash graphemes is not vertically centred with respect to the row for the 2-dash grapheme, but instead with respect to the upper half of the row, resulting in a cross rather than plus shape for the 2-dash grapheme.

This ruleset could now be applied to Morse code, which made it possible to create a font for CMC. This was made using an open-source program called FontForge (Williams, 2025). The resulting font table is shown in figure 8, with a sample text being shown in figure 1.

Fig 8. Font table for Condensed Morse Code in FontForge

Discussion

CMC, as a writing system and as a font has achieved its goals. It has achieved intuitiveness by preserving the Morse code representations in its glyphs, compactness by removing space redundancy, visual discernibility and ease of handwriting with its design, and consistency with the standard set of rules. Even so, there is no perfect writing system, and CMC is no exception. It takes more space than Compact Morse Code and could be more visually discernible if separate graphemes or glyphs were used for non-letter characters, as punctuation and numbers tend to “stick out” in writing. Another limitation is that the font does not consider prosigns and non-English Latin letters, such as some of the Letters in Polish or French. The prosign issue is largely unsolvable, as they are not part of the UTF standard, while the international Latin letters are an issue that can be solved in subsequent efforts. Lastly, is the issue that there are not, at the time of writing, alternative fonts for other Morse code standards, such as American Morse code. This can also be addressed in subsequent efforts, but additions to the dot combinations’ set of graphemes might be required to ensure the one-to-one correspondence between dot combinations and equivalent graphemes, for example in the case of “6” in American Morse code (· · · · ·).

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